GATE ME 90 Days Challenge

Day 73/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Thermodynamics + Applied Thermodynamics	**Thermo & Applied Thermo Revision**	**26 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on a full revision of Thermodynamics (pure substances, ideal-gas entropy, the second law with several reservoirs, the third law, throttling) and Applied Thermodynamics (Rankine and binary cycles, gas-turbine behaviour, multi-stage compressors, SI and CI engine performance, psychrometry, vapour-compression refrigeration and compounded steam turbines). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The third law of thermodynamics states that

(A) the entropy of every substance is zero at 0 K

(B) the entropy of a pure, perfectly crystalline substance is zero at absolute zero

(C) the entropy of an isolated system can never decrease

(D) absolute zero can be reached in a finite number of steps

Q2 NAT

A rigid tank contains dry saturated steam at 300 kPa (v_g = 0.6058 m³/kg). It is cooled until the pressure falls to 100 kPa, where v_f = 0.001043 m³/kg and v_g = 1.6941 m³/kg. The dryness fraction at the final state (round off to three decimal places) is _____

Q3 MCQ

Air (c_p = 1.005 kJ/kg·K, R = 0.287 kJ/kg·K) is compressed from 100 kPa, 300 K to 500 kPa, 450 K. The change in its specific entropy is about

(A) +0.407 kJ/kg·K

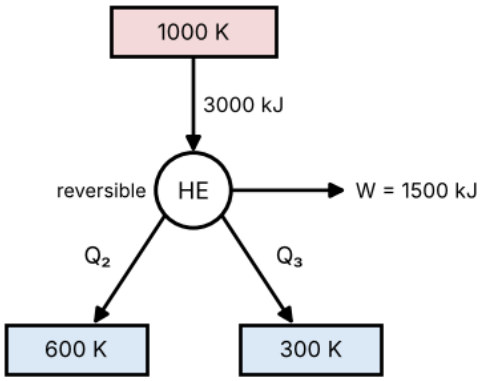
(B) −0.462 kJ/kg·K

(C) +0.054 kJ/kg·K

(D) −0.054 kJ/kg·K

Q4 NAT

A reversible engine receives 3000 kJ from a reservoir at 1000 K, exchanges heat with two other reservoirs at 600 K and 300 K, and produces 1500 kJ of work, as shown. The heat rejected to the 300 K reservoir (in kJ) is _____



Q5 MCQ

In a mercury–steam binary vapour cycle, mercury is used as the working fluid of the topping cycle mainly because

(A) its critical temperature is low

(B) its saturation pressure is low even at high temperatures, so heat can be added at a high temperature without very high pressures

(C) its latent heat of vaporisation is very large

(D) it does not need a condenser

Q6 NAT

In an ideal Rankine cycle, steam enters the turbine at 6 MPa and 500 °C (h = 3423.1 kJ/kg, s = 6.8826 kJ/kg·K) and the condenser pressure is 10 kPa. At 10 kPa: h_f = 191.81 kJ/kg, h_{fg} = 2392.1 kJ/kg, s_f = 0.6492 kJ/kg·K, s_{fg} = 7.4996 kJ/kg·K and v_f = 0.00101 m³/kg. The thermal efficiency of the cycle (in %, round off to one decimal place) is _____

Q7 MCQ

When the ambient (compressor inlet) temperature of a simple gas turbine plant with real (non-isentropic) compressor and turbine rises on a hot day, with the same pressure ratio and turbine inlet temperature,

(A) the net output and the thermal efficiency both increase

(B) the net output increases but the thermal efficiency falls

(C) the net output and the thermal efficiency both fall

(D) the net output falls but the thermal efficiency is unaffected

Q8 MCQ

Which one of the following is NOT an advantage of multi-stage compression with intercooling over single-stage compression to the same delivery pressure?

(A) Less work input per kg of air

(B) Higher volumetric efficiency of the first stage

(C) Lower delivery temperature in each stage

(D) Fewer moving parts and a simpler machine

Q9 NAT

A four-stroke SI engine of 2.0 L displacement runs at 3000 rpm with a volumetric efficiency of 85 % (based on an air density of 1.18 kg/m³). The air–fuel ratio is 14.7, the calorific value of the fuel is 44 000 kJ/kg and the brake thermal efficiency is 30 %. The brake power (in kW, round off to one decimal place) is _____

Q10 MCQ

The power output of an SI engine is controlled by ____, whereas that of a CI engine is controlled by ____.

(A) throttling the air–fuel mixture (quantity governing); varying the fuel quantity injected with unthrottled air (quality governing)

(B) varying the fuel quantity at a fixed air flow (quality governing); throttling the mixture (quantity governing)

(C) advancing the spark; retarding the injection

(D) changing the compression ratio; changing the injection pressure

Q11 NAT

Moist air at 25 °C and a total pressure of 101.325 kPa has a specific humidity of 0.012 kg/kg of dry air. The saturation pressure of water at 25 °C is 3.169 kPa. The degree of saturation of the air (round off to three decimal places) is _____

Q12 MCQ

In a vapour-compression refrigerator, the condenser temperature rises (on a hot day) while the evaporator temperature stays the same. Then

(A) the refrigerating effect per kg increases and the COP increases

(B) the refrigerating effect per kg is unchanged but the COP decreases

(C) the refrigerating effect per kg and the COP both decrease, and the compressor work per kg increases

(D) the compressor work per kg decreases and the COP increases

Q13 NAT

A room is kept at 25 °C. Its sensible heat load is 20 kW, and conditioned air is supplied at 14 °C. Taking the humid specific heat of the supply air as 1.022 kJ/kg·K (per kg of dry air), the required supply air flow (in kg of dry air per second, round off to two decimal places) is _____

Q14 NAT

A velocity-compounded (Curtis) impulse stage has two rows of moving blades. Steam leaves the nozzle at 800 m/s at 16° to the plane of the wheel. With symmetrical, frictionless blades, the blade speed for maximum diagram efficiency (in m/s, round off to one decimal place) is _____

Q15 MCQ

When wet steam at a high pressure is throttled to a low pressure (for example in a throttling calorimeter), its

(A) enthalpy stays the same and its dryness fraction increases (it may become superheated)

(B) enthalpy and dryness fraction both stay the same

(C) entropy stays the same and its dryness fraction decreases

(D) enthalpy decreases and its temperature stays the same

Answer Key

Q	Type	Answer
Q1	MCQ	(B) the entropy of a pure, perfectly crystalline substance is zero at absolute zero
Q2	NAT	0.357 (accept 0.354 to 0.360)
Q3	MCQ	(D) −0.054 kJ/kg·K
Q4	NAT	300
Q5	MCQ	(B) its saturation pressure is low even at high temperatures, so heat can be added at a high temperature without very high pressures
Q6	NAT	38.4 (accept 38.1 to 38.6)
Q7	MCQ	(C) the net output and the thermal efficiency both fall
Q8	MCQ	(D) Fewer moving parts and a simpler machine
Q9	NAT	45.0 (accept 44.6 to 45.4)
Q10	MCQ	(A) throttling the air–fuel mixture (quantity governing); varying the fuel quantity injected with unthrottled air (quality governing)
Q11	NAT	0.598 (accept 0.594 to 0.601)
Q12	MCQ	(C) the refrigerating effect per kg and the COP both decrease, and the compressor work per kg increases
Q13	NAT	1.78 (accept 1.76 to 1.79)
Q14	NAT	192.3 (accept 191.3 to 193.3)
Q15	MCQ	(A) enthalpy stays the same and its dryness fraction increases (it may become superheated)

Detailed Solutions

Q1 Answer: (B) the entropy of a pure, perfectly crystalline substance is zero at absolute zero

Concept: The third law fixes the zero of entropy: $S \to 0$ as $T \to 0$ for a pure substance in perfect crystalline equilibrium (Planck statement).

Glasses, mixtures and imperfect crystals keep a residual entropy at 0 K, so option A is too broad.

Option C is the increase-of-entropy principle (second law); the third law in fact implies that 0 K *cannot* be reached in a finite number of steps.

Answer: (B)

Exam tip: The third law is what makes *absolute* entropy values (used in combustion and chemical equilibrium) possible.

Q2 **Answer: 0.357 (accept 0.354 to 0.360)**

Concept: In a rigid closed tank the mass and the volume are fixed, so the specific volume stays constant during cooling.

$v_2 = v_1 = 0.6058$ m³/kg, which lies between v_f and v_g at 100 kPa, so the steam is wet.

$x_2 = (v - v_f)/(v_g - v_f) = (0.6058 - 0.001043)/(1.6941 - 0.001043) = 0.60476/1.69306$

$x_2 =$ **0.357**

Exam tip: On a T–v diagram a rigid-tank process is a vertical line; cooling dry steam drops it into the wet region.

Q3 **Answer: (D) −0.054 kJ/kg·K**

Concept: For an ideal gas, $\Delta s = c_p \ln(T_2/T_1) - R \ln(p_2/p_1)$.

Temperature term: $1.005 \ln(450/300) = 1.005 \times 0.4055 = 0.4075$ kJ/kg·K

Pressure term: $0.287 \ln 5 = 0.287 \times 1.6094 = 0.4619$ kJ/kg·K

$\Delta s = 0.4075 - 0.4619 =$ **−0.054 kJ/kg·K**

Exam tip: A negative Δs is allowed here: the compressor is not adiabatic in this data (heat leaves the air), so entropy can fall.

Q4 **Answer: 300**

Concept: For a reversible engine both laws apply as equalities: $\Sigma Q = W$ and $\Sigma Q/T = 0$ (Clausius).

First law: $Q_2 + Q_3 = 3000 - 1500 = 1500$ kJ (heat rejected at 600 K and 300 K)

Second law: $3000/1000 = Q_2/600 + Q_3/300 \rightarrow 3 = Q_2/600 + (1500 - Q_2)/300 = 5 - Q_2/600$

$Q_2 = 1200$ kJ and $Q_3 =$ **300 kJ** (both positive, so both are rejections, as assumed)

Exam tip: If a solution gives a negative Q, that reservoir is actually supplying heat; the sign tells you the direction.

Q5 **Answer: (B) its saturation pressure is low even at high temperatures, so heat can be added at a high temperature without very high pressures**

Concept: Rankine efficiency rises with the mean temperature of heat addition, but steam needs enormous pressures to boil at high temperatures.

Mercury boils at about 500 °C at only about 10 bar, so the topping cycle adds heat at a high temperature at a modest pressure.

Its condenser rejects heat to boil water for the steam (bottoming) cycle. Its latent heat is actually small, so a large mass of mercury must circulate.

Answer: (B)

Exam tip: Binary cycles try to make the overall cycle look like a Carnot cycle by stacking fluids that suit different temperature ranges.

Q6 **Answer: 38.4 (accept 38.1 to 38.6)**

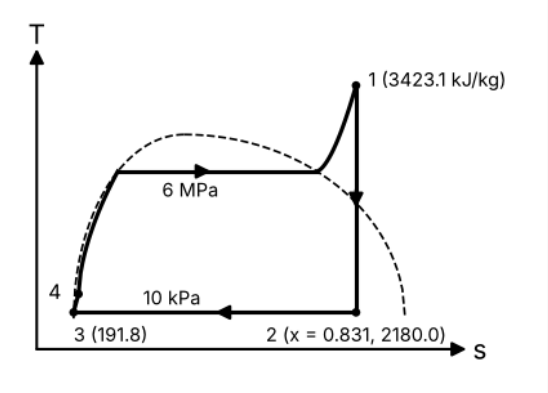
Concept: Ideal Rankine: isentropic turbine and pump; $\eta = (W_T - W_P)/Q_{in}$ (cycle shown below).

$x_2 = (6.8826 - 0.6492)/7.4996 = 0.8312$; $h_2 = 191.81 + 0.8312 \times 2392.1 = 2180.0$ kJ/kg; $W_T = 1243.1$ kJ/kg

$W_P = v_f \Delta p = 0.00101 \times (6000 - 10) = 6.05$ kJ/kg; $h_4 = 197.86$ kJ/kg

$Q_{in} = 3423.1 - 197.86 = 3225.2$ kJ/kg; $\eta = (1243.1 - 6.05)/3225.2 =$ **38.4 %**

Exam tip: The exhaust dryness here is only 0.83; below about 0.88 blade erosion becomes a concern, which is why reheat is used.



Q7 **Answer: (C) the net output and the thermal efficiency both fall**

Concept: Compressor work per kg is proportional to the inlet temperature, while the turbine work depends on the (unchanged) turbine inlet temperature.

A hotter intake raises the compressor work and lowers the air density, so the mass flow falls: the net output drops.

With real (non-ideal) machines the efficiency also falls, because the compressor work rises while the heat added ($T_3 - T_2$) shrinks less than the net work.

Answer: (C)

Exam tip: This is why inlet-air cooling (evaporative or chiller) is used on gas turbines in hot climates.

Q8 **Answer: (D) Fewer moving parts and a simpler machine**

Concept: Staging with intercooling brings the compression closer to isothermal and reduces the pressure ratio per stage.

Work input falls (shaded area saved on the p–V diagram), each stage delivers cooler air, and the lower stage pressure ratio raises the volumetric efficiency.

The price is more cylinders, valves and an intercooler: the machine becomes more complex.

Answer: (D)

Exam tip: Optimum intermediate pressure for two stages is $\sqrt{(p_1 p_3)}$, which gives equal work in both stages.

Q9 **Answer: 45.0 (accept 44.6 to 45.4)**

Concept: Air flow = $\eta_v \rho_a V_s$ (N/2) for a four-stroke engine; fuel flow = air flow/(A/F); BP = $\eta_{bth} \dot{m}_f$ CV.

$\dot{m}_a = 0.85 \times 1.18 \times 0.002 \times (3000/120) = 0.05015$ kg/s

$\dot{m}_f = 0.05015/14.7 = 0.003412$ kg/s

BP = $0.30 \times 0.003412 \times 44\,000$ = **45.0 kW**

Exam tip: In a four-stroke engine each cylinder breathes once every two revolutions; forgetting N/2 doubles the answer.

Q10 **Answer: (A) throttling the air–fuel mixture (quantity governing); varying the fuel quantity injected with unthrottled air (quality governing)**

Concept: An SI engine must burn a near-stoichiometric mixture for reliable flame propagation, so it is throttled: less mixture, same ratio (quantity governing).

A CI engine draws unthrottled air and the load is set by how much fuel is injected, so the A/F ratio varies (quality governing).

Throttling losses (pumping work) are one reason SI engines are less efficient at part load.

Answer: (A)

Exam tip: 'Quantity' and 'quality' refer to the mixture: its amount (SI) or its strength (CI).

Q11 **Answer: 0.598 (accept 0.594 to 0.601)**

Concept: Degree of saturation $\mu = \omega/\omega_s$, where ω_s is the specific humidity of saturated air at the same temperature and total pressure.

$\omega_s = 0.622 \times 3.169/(101.325 - 3.169) = 0.020081$ kg/kg

$\mu = 0.012/0.020081$ = **0.598**

For comparison, $p_v = \omega p/(0.622 + \omega) = 1.918$ kPa, so $\varphi = 1.918/3.169 = 0.605$: μ is slightly smaller than φ.

Exam tip: $\mu = \varphi(p - p_s)/(p - p_v)$; the two are nearly equal at room conditions and both equal 1 for saturated air.

Q12 **Answer: (C) the refrigerating effect per kg and the COP both decrease, and the compressor work per kg increases**

Concept: On the p–h diagram, a higher condensing pressure moves the saturated-liquid point (and the throttled state) to a higher enthalpy, and lengthens the compression line.

Refrigerating effect $h_1 - h_4$ falls because $h_4 = h_3$ rises; compressor work $h_2 - h_1$ rises.

So COP = $(h_1 - h_4)/(h_2 - h_1)$ falls on both counts.

Answer: (C)

Exam tip: Keep condensers clean and well ventilated: every degree of extra condensing temperature costs roughly 2–3 % in COP.

Q13 **Answer: 1.78 (accept 1.76 to 1.79)**

Concept: The supply air absorbs the room sensible heat as it warms from the supply temperature to the room temperature: RSH = $\dot{m}_a\, c_{pm}\, (t_{room} - t_{supply})$.

$\dot{m}_a = 20/(1.022 \times (25 - 14)) = 20/11.242$

$\dot{m}_a$ = **1.78 kg/s**

Exam tip: The latent load does not enter this sizing; it fixes how dry the supply air must be (the RSHF line on the chart).

Q14 **Answer: 192.3 (accept 191.3 to 193.3)**

Concept: For an impulse stage with n rows of moving blades (symmetrical, frictionless), the optimum blade-speed ratio is $u/V_1 = \cos\alpha/(2n)$ and the maximum diagram efficiency is $\cos^2\alpha$.

Two rows: $u/V_1 = \cos 16°/4 = 0.9613/4 = 0.2403$

$u = 0.2403 \times 800$ = **192.3 m/s** (maximum diagram efficiency $\cos^2 16°$ = 92.4 %)

Exam tip: Curtis stages handle a large enthalpy drop at a low blade speed, but their actual efficiency falls quickly as rows are added.

Q15 **Answer: (A) enthalpy stays the same and its dryness fraction increases (it may become superheated)**

Concept: Throttling is adiabatic with no work and negligible KE change, so $h_1 = h_2$; it is irreversible, so entropy rises.

At the lower pressure h_f is much smaller, so the same enthalpy corresponds to a larger dryness fraction; if h exceeds h_g at the low pressure, the steam becomes superheated.

On the h–s (Mollier) chart throttling is a horizontal line moving to the right.

Answer: (A)

Exam tip: A throttling calorimeter works only for nearly dry steam (x above about 0.95); wetter steam needs a separating calorimeter first.

What next?

Day 73 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

Next, Day 74/90: Materials & Processes Revision. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.